



电子科技大学
格拉斯哥学院
Glasgow College, UESTC

ECECS

CS 15
Academic Writing

WORLD
CHANGING
GLASGOW

Aims

In this CS you will

- Consider what makes a text academic
- Analyse features of an academic text
- Examine ‘evaluative’ language
- Examine report title

1. Academic Writing

Task 1.1 What makes a text academic

- answers or explores a specific question
- uses well-structured, logical argument, based on research by writer or previous writers/researchers
- refers to previous research (cited within text and referenced at the end of text)
- tends to use an impersonal/objective **style**. Uses specific language features to achieve this style:
 - Avoids personal & contracted language forms
 - Has high lexical density & nominalisation (noun phrases)
 - Uses impersonal language forms (e.g. passive), but not always!
 - Use of modals and other tentative language

nominalisation

1. We **analysed** the data from the experiment and it revealed that children **react** when they **have** too much sugar.
2. The **analysis** of the data revealed children's **reaction** to excessive sugar **intake**.

Task 1. 2 Cohesion and coherence

Cohesion

The **micro** level of the text, i.e. the words and sentences and how they join together (e.g., reference words, linking devices)

Coherence

The **macro** level features of a text , i.e. organisation and connection of ideas and whether they can be understood by the reader(e.g., topic sentences, thesis statement, the summary in the concluding paragraph.

It can be improved by using an outline before writing, or check that the ideas are logical and well organised

Academic writing

Coherence & Cohesion

Good written communication:

- Clarity (clear)
- Precision (accurate)
- Conciseness (brief)
- relevance
- good organisation
- logical order
- flow within and between sentences + paragraphs
- Unity (main point-supporting details)

Aspects to cover in this session

--text analysis of waste management

1. The principle of Given-New information
2. Creating flow with repetition
3. Grammatical referencing (pronouns)
4. Evaluative language

Task 1.3 The principle of Given-New information

- Using a Given-New principle helps the reader understand how a new piece of information fits into information they already know.
- Orientation first!
- What 2 main types of 'given information' are there likely to be in a text?
- 1. information **already mentioned** in the text (not always in the exact same words).
- 2. information **already familiar to readers** with general or specialist knowledge of the subject

Paragraph 5

- Incineration as an alternative approach to landfill is an option that is being increasingly employed globally [7], [8]. According to [8], burning waste can significantly reduce the amount of solid waste that would otherwise be put into landfill sites by 90%. Another significant benefit of incineration is that it enables the production of Waste-to-Energy (WTE), where electricity can be generated from the heat produced [6], [7], [8]. The Manila metropolitan area reported an average waste disposal weight of 10 kilotons daily in 2020. [6] calculates that this could provide the city with a potential 235 764 MW of electricity every day, thereby enabling dependence on fossil fuel use for power generation to be decreased.

Given – new information (paragraph 5)

- **Incineration** as an alternative approach to landfill is an option that is being increasingly employed globally [7], [8]. According to [8], **burning waste** **can significantly reduce the amount of solid waste that would otherwise be put into landfill sites by 90%.**
- *The term 'burning waste' is given (called 'incineration' in the previous sentence) and then the new information (in yellow) about it comes later in the sentence i.e. the fact it can significantly reduce the amount of solid waste.*

Principle: fitting new into info already given (known)

Task 1.3 Identifying Given-New information (paragraph 5)

- Incineration as an alternative approach to landfill is an option that is being increasingly employed globally [7], [8]. According to [8], burning waste can significantly reduce the amount of solid waste that would otherwise be put into landfill sites by 90%. Another significant benefit of incineration is that it enables the production of Waste-to-Energy (WTE), where electricity can be generated from the heat produced [6], [7], [8]. The Manila metropolitan area reported an average waste disposal weight of 10 kilotons daily in 2020. [6] calculates that this could provide the city with a potential 235 764 MW of electricity every day, thereby enabling dependence on fossil fuel use for power generation to be decreased.

Task 1.4 Creating flow with repetition

- **synonymous repetition**

A. Use identical words

B. Use of synonyms

C. Use of umbrella terms

Task 1.4 A. Using Identical Words (paragraph 3)

- A significant danger of unregulated landfill sites relates to the suitability of the site and the uncontrolled way in which waste can accumulate [4]. If the ground on which the landfill is sited lacks stability in the first instance, then the stability of the landfill that accumulates above it will also be unstable [4]. There is an additional issue that piled waste may not be adequately compacted when it is put in the waste pile. Both of these issues can be a contributing factor to “slope failure” [4], when the accumulated pile of landfill can give way, sometimes with catastrophic consequences and loss to human life. The latter was found to be a major cause of the July 2000 incident in the Payatas landfill site in the Quezon City suburb of the Manila metropolitan area. Here, in excess of 1 million m³ of waste collapsed [4], crushing more than more than 100 homes in the site vicinity and causing an estimated 300 fatalities [3].

Task 1.4 A. Using Identical Words (paragraph 3)

- A significant danger of unregulated landfill sites relates to the suitability of the site and the uncontrolled way in which waste can accumulate [4]. If the ground on which the landfill is sited lacks stability in the first instance, then the stability of the landfill that accumulates above it will also be unstable [4]. There is an additional issue that piled waste may not be adequately compacted when it is put in the waste pile. Both of these issues can be a contributing factor to “slope failure” [4], when the accumulated pile of landfill can give way, sometimes with catastrophic consequences and loss to human life. The latter was found to be a major cause of the July 2000 incident in the Payatas landfill site in the Quezon City suburb of the Manila metropolitan area. Here, in excess of 1 million m³ of waste collapsed [4], crushing more than more than 100 homes in the site vicinity and causing an estimated 300 fatalities [3].

Use of definite article: the + noun

Refer back an items in text to avoid ambiguity, unlike pronouns

B: synonym (from paragraph 3)

- Both of these issues can be a contributing factor to “slope failure” [4], when the accumulated pile of landfill can give way, sometimes with catastrophic consequences and loss to human life. The latter was found to be a major cause of the July 2000 incident in the Payatas landfill site in the Quezon City suburb of the Manila metropolitan area. Here, in excess of 1 million m³ of waste collapsed [4], crushing more than 100 homes in the site vicinity and causing an estimated 300 fatalities [3].

B: synonym (from paragraph 3)

- Both of these issues can be a contributing factor to “slope failure” [4], when the accumulated pile of landfill can **give way**, sometimes with catastrophic consequences and **loss to human life**. The latter was found to be a major cause of the July 2000 incident in the Payatas landfill site in the Quezon City suburb of the Manila metropolitan area. Here, in **excess of** 1 million m³ of waste **collapsed** [4], crushing **more than** 100 homes in the site vicinity and causing an estimated 300 **fatalities** [3].

Show variation of language but ensure accurate synonyms

C: Umbrella terms (paragraph 2)

- The Philippines has faced issues with controlling solid waste disposal in recent decades. In many countries, particularly in developing nations like The Philippines, the majority of waste is disposed of in solid waste landfills that are often not subject to strict controls and regulations [4]. This approach to waste disposal can result in significant health and safety risks for the local communities who live nearby landfills [3], and pose environmental and pollution risks [2], such as groundwater supply contamination [5]. Despite government regulations introduced in The Philippines in 2001 to better control waste management, the amount of solid waste generation there has not been curbed to date. This is coupled with a continued shortage of alternative waste disposal infrastructure [6].

C: Umbrella terms (paragraph 2)

- The Philippines has faced **issues** with controlling solid waste disposal in recent decades. In many countries, particularly in developing nations like The Philippines, the majority of waste is disposed of in solid waste landfills that are often not subject to strict controls and regulations [4]. This **approach** to waste disposal can result in significant health and safety risks for the local communities who live nearby landfills [3], and pose environmental and pollution risks [2], such as groundwater supply contamination [5]. Despite government regulations introduced in The Philippines in 2001 to better control waste management, the amount of solid waste generation there has not been curbed to date. This is coupled with a continued shortage of alternative waste disposal **infrastructure** [6].

Describe a class of objects/things; simplify expressions

Task 1.5 Using pronouns (and other referencing devices) – Paragraph 6

- However, incineration is not without its risks. When solid waste is incinerated, it creates a hazardous by-product called fly-ash, which contains significant amounts of heavy metals, chlorides and dioxins [8]. These toxic by-products of incineration have long been known to cause adverse health effects in humans such as potentially leading to various types of cancer and have even been reported to lower reproductive capabilities [7]. These toxic particles can affect people either through inhalation or food consumption where food has been grown or stored nearby incinerators. It can also potentially affect people further afield when this food is transported [7].

Task 1.5 Using pronouns (and other referencing devices) – Paragraph 6

- However, incineration is not without **its** risks. When solid waste is incinerated, **it** creates a hazardous by-product called fly-ash, which contains significant amounts of heavy metals, chlorides and dioxins [8]. **These** toxic by-products of incineration have long been known to cause adverse health effects in humans such as potentially leading to various types of cancer and have even been reported to lower reproductive capabilities [7]. These toxic particles can affect people either through inhalation or food consumption where food has been grown or stored nearby incinerators. **It** can also potentially affect people further afield when this food is transported [7].

Make a text more concise; Help with bridging and flow throughout text

Use of pronouns to replace given terms

Different types:

- Personal pronouns eg *which they claim... (not in this text)*
- Bald demonstrative pronouns eg *this is coupled by, this could provide*
- Possessive adjectives eg *their respective systems (not in this text)*
- This/These + repetition of given term eg *these toxic by-products, these pollutants*
- This/These + umbrella term eg *this approach*

2. Content and language in your evaluation section

Content:

- No language of personal opinion
- Include benefits and challenges/limitations
- Include a balanced conclusion based on the evaluation

Language:

- Use of cautious language (especially modal verbs in this text)
- Lexical words which are not definitive
- Use of noun phrases to make the text concise and precise
- Strong words for emphasis

01

Content features



Building a solid foundation for your arguments.

1. Avoid Subjective Opinion

Use evidence-based evaluation instead of personal feelings.

✗ Subjective Opinion

I think incineration is way better than landfill, and it's obvious everyone should use it.

✓ Academic Evaluation

Based **on the data** from [8], incineration can reduce landfill volume by **90%** while generating electricity, though it presents significant environmental risks.

2. Include Benefits & Challenges/limitations

Present a balanced view by addressing both sides.



Benefit

Incineration offers the advantage of waste-to-energy production, which can reduce reliance on fossil fuels.



Challenge / Limitation

However, the process produces toxic fly ash containing heavy metals and dioxins, which pose long-term health risks to nearby communities.

3. Include a Balanced Conclusion based on evaluation

Draw a conclusion based on your evaluation.

EXAMPLE CONCLUSION

Overall, while incineration can effectively reduce landfill volume and produce renewable energy, its high financial cost and persistent toxic emissions mean it cannot be considered a universally safe replacement for landfill without further technological improvement.

02

Language Features

Refining your expression for precision.

1. Use of Cautious Language

Use modal verbs to show different levels of certainty.



Possibility

These toxic particles **can** affect people either through inhalation or food consumption



Potential Impact

It can also potentially affect people further afield when this food is transported



Uncertainty

it is difficult to know to what extent these newer incineration technologies **may** prove to be a safer alternative

2. Non-Definitive Words

See how these words work together in a sentence.

Studies **suggest** that fly ash **may** contain significant amounts of dioxins, which are **potentially** harmful to human health.

suggest

Indicates the direction of research findings,
not a final conclusion.

may

a possibility, **not an absolute fact**

Potentially

Emphasizes potential risk or impact, **not a confirmed harm.**

2. Non-Definitive Lexical Words

Choose words that reflect the tentative nature of research.



Avoid: Definitive Words

always · never · definitely
prove · certain

These words claim absolute truth, which is rare
in academic research.



Use: Non-Definitive Words

suggest · indicate · potential
possible · likely · tend to

These words reflect the tentative, exploratory
nature of scientific findings.

3. Use of Noun Phrases

Make your writing concise and precise.

✗ Wordy Version

When you burn waste, it creates by-products that are hazardous.

✓ Concise Noun Phrase

The **incineration of solid waste** produces **hazardous by-products** including **toxic fly ash** containing **heavy metals** and **dioxins**.

4. Strong Words for Emphasis

Highlight key points with powerful vocabulary.

These **toxic** by-products have **long been known** to cause **adverse health effects** in humans, such as **potentially leading to** various types of cancer and even **lowering reproductive capabilities**.

toxic, adverse

Emphasize the **severity of harm**.
these words create a strong
emotional impact.

long been known

Highlights the **long-term and widespread nature of awareness**. It
implies that the risks are not new
discoveries but have been
documented for a long time.

potentially leading to

Indicates a **causal relationship**
between the substance and its
outcomes

Task 2 Evaluative language (paragraph 6)

- However, incineration is not without its risks. When solid waste is incinerated, it creates a hazardous by-product called fly-ash, which contains significant amounts of heavy metals, chlorides and dioxins [8]. These toxic by-products of incineration have long been known to cause adverse health effects in humans such as potentially leading to various types of cancer and have even been reported to lower reproductive capabilities [7]. These toxic particles can affect people either through inhalation or food consumption where food has been grown or stored nearby incinerators. It can also potentially affect people further afield when this food is transported [7]. And even though more recent innovations in incineration technology may be able to mitigate these effects, these pollutants are still not eliminated. It is also worth noting that since there may be a delayed onset of any adverse effects from these toxins in humans, it is difficult to know to what extent these newer incineration technologies may prove to be a safer alternative to landfill disposal [7]. [8] also notes the significant financial cost of building incinerator plants with the latest technological developments. There is also the issue that people who previously depended on solid landfill sites for their livelihoods can no longer profit from this [3].

Task 3. Report **title** evaluation

What makes a good report title?

- 1.Explaining a survey related to making aspects of Internet of Things secure
- 2.Internet of Things Security: A Survey

Task 3. Report **title** evaluation

What makes a good report title?

Explaining a survey related to making aspects of Internet of Things secure

Internet of Things Security: A Survey 

- Concise (3-10 words)
- Informative (summarize the topic)
- Upper case for main content words (not grammar words)
- No full stop
- Use of noun phrases
- Use colon to shorten or simplify

Task 3. Report **title** evaluation

1. Healthcare IoT Devices: Security and Privacy Framework
2. A description of the greatest things achieved in grand challenges in engineering in the 21st century.
3. The challenge(s) relating to the design of emergency communication systems which can be used in regions whose communication systems have been damaged by natural disasters.
4. Global Grand Challenges for International Development

Task 3.1 Report **title** evaluation

1. Healthcare IoT Devices: Security and Privacy Framework ☒
2. A description of the greatest things achieved in grand challenges in engineering in the 21st century.
3. The challenge(s) relating to the design of emergency communication systems which can be used in regions whose communication systems have been damaged by natural disasters.
4. Global Grand Challenges for International Development ☒

Task 3.1 Report **title** evaluation

1. A description of the greatest things achieved in grand challenges in engineering in the 21st century.

21st Century Engineering Grand Challenges Achievements

2. The challenge(s) relating to the design of emergency communication systems which can be used in regions whose communication systems have been damaged by natural disasters.

Natural Disasters and Emergency Communication System Design

Task 3.1 Report title evaluation

1. Internet of Things Security: A Survey
2. Healthcare IoT Devices: Security and Privacy Framework
3. 21st Century Engineering Grand Challenges Achievements
4. Natural Disasters and Emergency Communication System Design
5. Global Grand Challenges for International Development



University
of Glasgow

Thank you!

#UofGWorldChangers



@UofGlasgow